Day 3 afternoon — Stan and diagnostics

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Thursday 12 November 2026 · Pontoppidanstræde 101, room 1.001

Stan program blocks · cmdstanr · HMC and NUTS · R-hat, ESS, MCSE, divergences · computation checks versus model checks

Today

08:45	**Lecture A**	Monte Carlo error · Markov chains · the Metropolis algorithm
09:45	**Exercise A**	Monte Carlo error in R · run and read a Metropolis sampler · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Stan: model blocks, HMC and NUTS, chains, diagnostics
13:30	**Exercise B**	Run Stan, check against the exact answer, read diagnostics · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

Follow-up from the morning plenary

Open points from Peer review A

Fill in during lunch: the one or two questions the plenary did not settle.

1. …
2. …

By the end of this lecture you can

1. **Read** a Stan program block by block
2. **Run** it from R with cmdstanr and extract the draws
3. **Explain** in words what HMC and NUTS do differently from Metropolis
4. **Check** a fit: R-hat, ESS, MCSE, divergences, treedepth, E-BFMI
5. **Separate** checking the computation from checking the model

What Stan does for you

This morning, by hand

You wrote log(likelihood × prior).

You chose the step size.

One chain; you judged it by eye.

With Stan

You write the model; Stan builds log(likelihood × prior).

Stan tunes itself during warmup.

Four chains and the diagnostics, by default.

- Stan's sampler is HMC with NUTS — a smarter way to propose than a random walk.
- It still produces a Markov chain: draws are dependent, warmup is discarded, ESS and MCSE still apply.

A Stan program has blocks

```
data {
  int<lower=0> n;            // trials
  int<lower=0, upper=n> y;   // successes
  real<lower=0> a;           // prior: Beta(a, b)
  real<lower=0> b;
}
parameters {
  real<lower=0, upper=1> theta;
}
model {
  theta ~ beta(a, b);        // prior
  y ~ binomial(n, theta);    // likelihood
}
generated quantities {
  int y_next10 = binomial_rng(10, theta);
}
```

block	what goes in
data	everything you know: observations and prior settings
parameters	the unknowns, with their allowed range
model	prior and likelihood
generated quantities	computed once per draw: predictions, derived quantities

The file from Day 2:
day-2/demos/models/beta_binomial.stan

Running it from R

```
library(cmdstanr)
# compile once
mod <- cmdstan_model("models/beta_binomial.stan")

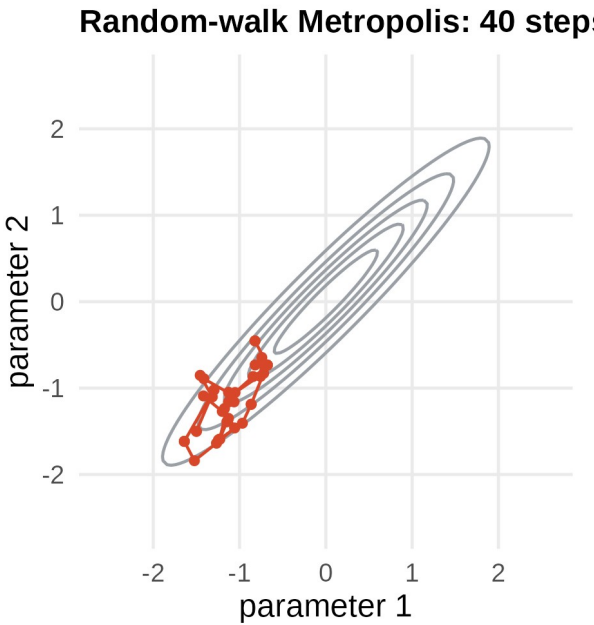
fit <- mod$sample(
  data   = list(n = 20, y = 14, a = 2, b = 2),
  chains = 4, seed = 2026
)

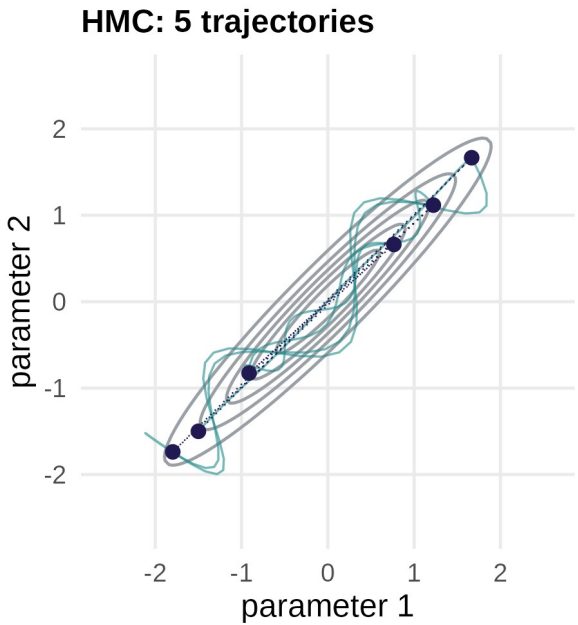
fit$summary()              # summaries + R-hat, ESS
fit$diagnostic_summary()   # divergences, treedepth, E-BFMI

draws <- fit$draws(format = "df")   # one row per draw
mean(draws$theta > 0.6)
```

- Compiling turns the model into a program: takes a minute, once.
- Defaults: 4 chains × (1,000 warmup + 1,000 kept).
- The prior's numbers a, b are passed as data: to try another prior, change a = 2, b = 2 in the data list. No recompiling.
- draws: one row per draw, one column per quantity — use it like rbeta() output.

HMC: proposals that follow the shape





- Think of a ball rolling on a landscape whose valleys are high-posterior regions.
- Give it a random push; follow its path for a while; stop: that is the proposal.
- The slope (gradient) steers the path along the posterior, so it travels far and is still accepted.

40 gradient-free steps vs 5 long, guided moves: HMC explores far more per draw.

NUTS and warmup

NUTS (No-U-Turn Sampler)

HMC needs a path length. NUTS chooses it for each draw: it extends the path until it starts to turn back.

The tree depth counts how many times the path was doubled.

Warmup (default 1,000 per chain)

Stan tunes the step size and the scale of each parameter.

Warmup draws are not used for inference.

Tuned settings are then fixed for the kept draws.

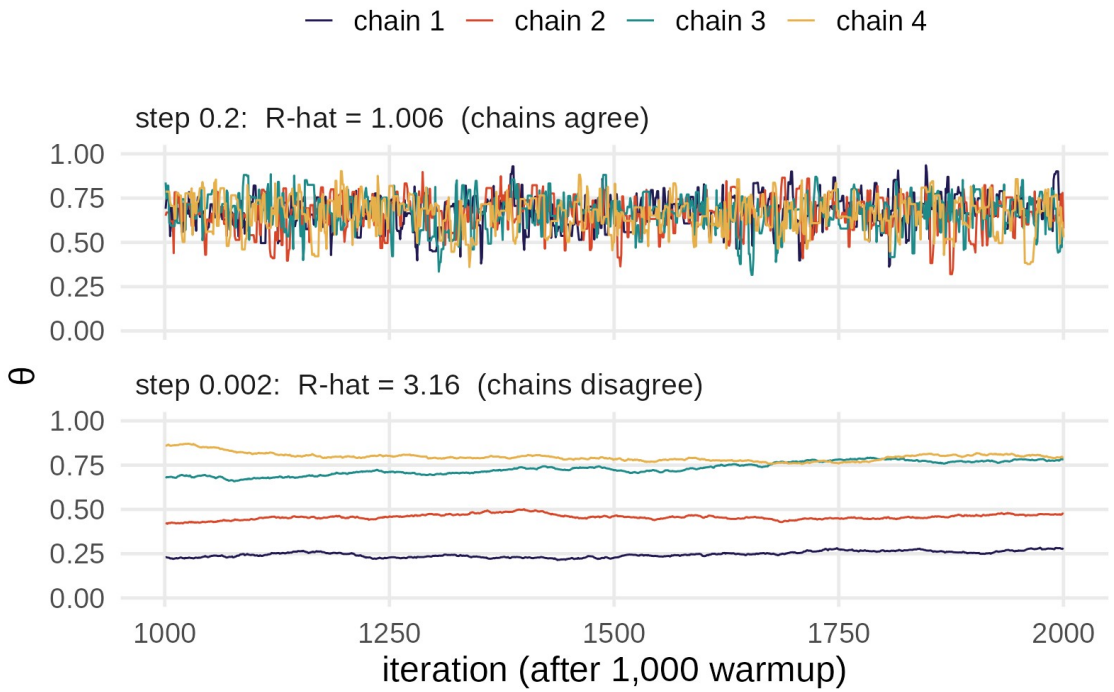
- Different chains start from different random values.
- Kept draws are still dependent — but usually much less than random-walk draws.

Six numbers to check after every fit

diagnostic	question it answers	look for
R-hat	Do the chains agree with each other?	< 1.01
bulk ESS / tail ESS	Enough information for the centre / the tails (intervals)?	> 400
MCSE	How precise is this computed number?	small vs the posterior SD
divergences	Did the paths fail in a difficult region?	0
max treedepth hits	Were paths cut off before they finished?	0
E-BFMI	Did the chains explore the energy levels well?	> 0.3

Thresholds are screening rules, not proofs. A warning means: investigate before you interpret.

R-hat: do the chains agree?

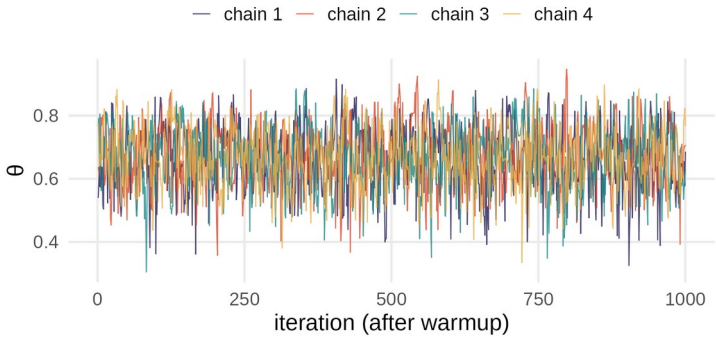


- R-hat compares the variation between chains with the variation within chains.
- Same distribution in every chain → R-hat ≈ 1.
- Chains stuck in different places → R-hat ≫ 1.
  - Here: this morning's sampler, four chains, two step sizes.

Reading a real Stan summary: the bus

variable	mean	sd	mcse_mean	5%	95%	rhat	ess_bulk	ess_tail
theta	0.668	0.094	0.0025	0.507	0.817	1.00	1404	1735
y_next10	6.71	1.76	0.035	4	9	1.00	2546	3243

```
divergences per chain:    0 0 0 0
max treedepth hits:       0 0 0 0
E-BFMI per chain:         1.21 1.00 1.21 1.20
```



- All checks pass → we may interpret the numbers.
- Exact answer: mean 0.667, 90% interval 0.504 – 0.814 ✓
- ESS 1,404 of 4,000 draws; MCSE 0.0025, tiny compared with the posterior SD 0.094.

Divergences: a warning you must not ignore

What it means

The simulated path flew off the posterior where the shape bends too sharply. That region is under-explored, so summaries can be biased.

When you see it

Typical in hierarchical models (Day 4): a “funnel” between a group-level SD and the group effects.

What to do

- Check the model code, data and parameter scales first: is something wrong?
- Rewrite the model so the shape is easier (reparameterisation, e.g. non-centred — Day 4).
- Smaller steps: adapt_delta = 0.95 or 0.99 can help, but is not a cure. More draws do not fix it.

Two different checks

Computation check

Question: did we compute THIS posterior correctly?

Tools: R-hat, ESS, MCSE, divergences, comparing with an exact answer.

Fix: more draws, better parameterisation, fix bugs.

Model check

Question: is THIS model good enough for the data and the question?

Tools: prior and posterior predictive checks, a different prior (sensitivity).

Fix: change the model.

Perfect diagnostics can describe a poor model very precisely. Passing one check says nothing about the other.

Worked example: computation or model?

- a) R-hat for one parameter is 1.08.
- b) Simulated data from the posterior never show as many zeros as the real data.
- c) Three divergences out of 4,000 draws.
- d) With prior Beta(10, 10) instead of Beta(2, 2), the posterior mean moves from 0.67 to 0.60.
- e) $P(\theta > 0.6) = 0.76$ with MCSE 0.02, but you need to know whether it is above 0.75.

For each: computation or model? What would you do next?

Worked example: answers

	type	next step
a) R-hat 1.08	computation	do not interpret; look at the traces, run longer or fix the model
b) too few zeros	model	the model misses a feature of the data: change the model
c) 3 divergences	computation	even a few matter: check the code, reparameterise
d) prior Beta(10,10)	model (sensitivity)	report that the conclusion depends on the prior; exact Beta(24,16): 0.600, Stan 0.602
e) MCSE 0.02	computation	$0.76 \pm 2 \times 0.02$ includes 0.75: run more draws ($4 \times$ draws $\rightarrow$ half the MCSE)

Exercise B · 13:30 – 15:00

- Read the supplied Stan model: find data, parameters, prior, likelihood, generated quantities.
- Fit it with cmdstanr and compare with the exact beta posterior.
- Read the diagnostics and write down whether the fit can be trusted.
- Change the prior (only the data list) and compare.
- Write your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

15:00

peer review B

If Stan does not compile on your laptop: continue on UCloud or a group member’s laptop. New roles this block.

Peer review B · 15:00 – 15:45

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

2 min

Wrap-up

Points that need longer go to tomorrow's follow-up.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

Daily diary · 15:45 – 16:00

menti.com · code: ____ ____

1. Three things I have learned today
2. Three things I am not yet very familiar with